

Payment & Financial Services - Document 7 - Page 1/5

Document ID: PAY-007 | Classification: Internal Use

Payment user authentication: 3DS2 flows, biometric verification, and risk-based authentication.

Merchant KYC: business registration, bank account verification, beneficial owner disclosure.

Error code 4001: Invalid payment method. Verify card details and expiry. Support 3DS if required.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 7 - Page 2/5

Document ID: PAY-007 | Classification: Internal Use

Payment transaction logging: structured logs, trace IDs, and observability for incident investigation.

Subscription billing: prorate on upgrade/downgrade. Credit unused portion. Send invoice 7 days before charge.

Error code 4002: Insufficient funds. Customer must use alternative payment method or add funds.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 7 - Page 3/5

Document ID: PAY-007 | Classification: Internal Use

Payment service deployment: blue-green releases, canary deployments, and rollback procedures.

Multi-currency: store amount in minor units (cents). Display in major units. Use ISO 4217 codes

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EU payment failures: Verify PSD2 compliance, SCA completion, and issuer 3DS configuration.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging.
For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
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Payment & Financial Services - Document 7 - Page 4/5

Document ID: PAY-007 | Classification: Internal Use

Billing and invoicing: subscription management, proration, tax calculation, and invoice generation.

Fraud score > 80: block transaction. Score 50-80: require 3DS. Score < 50: auto-approve.

Intermittent payment failures: Review gateway logs, check rate limits, verify certificate expiry.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 7 - Page 5/5

Document ID: PAY-007 | Classification: Internal Use

Refund and chargeback processing: initiation, status tracking, dispute resolution, and reason codes.

Audit log retention: 7 years for financial transactions. Immutable storage. Tamper-evident.

Currency mismatch: Ensure amount and currency code match. INR to USD conversion requires FX rates.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.